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Assignment -3

- Title
Subnetting

- Aim
Subnetting (Language : C / C++ / Python / Java) / Packet tracer)

- Objectives
1. To understand and learn the concept of IP address, subnet mask and subnetting.

- Theory

1. Internet Protocol
- IPV4 : 34-bit address written in dotted decimal notation.
Example : 192.168.1.10
- IPV6 : 128-bit address written in hexadecimal
Example : 2001 : db8 : : 1

2. Classful Addressing.

Class	Range	Default subnet Mask	Use
A	1-126	255.0.0.0 (18)	Large networks
B	128-191	255.255.0.0 (116)	Medium networks
C	192-223	255.255.255.0 (124)	Small networks
D	224-239	-	Multicast
E	240-255	-	Experimental

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3. Networks and hosts per subnet
Number of subnets = 2^n
Hosts per subnet = $2^h - 2$
Where; n = borrowed bits
h = remaining host bits
Example: 192.168.1.0/26
Host bits = 6
Hosts per subnet:
$2^6 - 2 = 62$ hosts
Number of subnets from 124:
$2^2 = 4$ subnets.

4. Subnetting Example
Given,
Network: 192.168.1.0/24
Need 4 subnets.
Borrow 2 host bits:
/24 → /26
Subnet mask:
255.255.255.192

Subnet	Network Address	Host Range	Broad
1	192.168.1.0	.1 - .62	.6
2	192.168.1.64	.65 - .126	.12
3	192.168.1.128	.129 - .190	.19
4	192.168.1.192	.193 - .254	.25

- Student's Observation
Successfully divided a network into smaller subn
using subnetting.

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• FAQs

1) Describe classful and classless IP addressing schemes with examples.

(i) Classful - Uses fixed classes A, B, C with default masks
Example : 192.168.1.0/24

(ii) Classless - Uses CIDR notation and flexible prefix lengths
Example : 192.168.1.0/26

2) What are different special / reserved IPv4 addresses?

- 0.0.0.0 - Unspecified / default route
- 127.0.0.0/8 - Loopback
- 255.255.255.255 - Limited broadcast
- Private : 10.0.0.0/8
- Private : 172.16.0.0/12
- Private : 192.168.0.0/16

3) What is subnetting?

Subnetting is the process of dividing one large network into smaller networks called subnets.
Example : 192.168.1.0/24 → 4 subnets using /26
It improves IP address utilization, network management and security.

4) Define FLSM, VLSM and CIDR

FLSM : All subnets have the same size
VLSM : Subnets can have different sizes according to requirements.

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CIDR : Classless addressing using a prefix length
Eg : 124, 126.

14/8/26

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```
#include <iostream>
#include <sstream>
#include <string>
using namespace std;
int main() {
  string ip;
  cout << "Enter an IP address (format: x.x.x.x): ";
  cin >> ip;
  stringstream ss(ip);
  string octet;
  int firstOctet;
  getline(ss, octet, '.');
  firstOctet = stoi(octet);
  cout << endl;
  cout << "--- Results ---" << endl;
  cout << "IP Address      : " << ip << endl;
  if (firstOctet >= 1 && firstOctet <= 126) {
    cout << "Class          : A" << endl;
    cout << "Subnet Mask     : 255.0.0.0" << endl;
    cout << "No. of Networks  : 2^7 (128 possible networks)" << endl;
    cout << "Usable Hosts    : 2^24 - 2 (16,777,214)" << endl;
    cout << "Starting IP     : " << firstOctet << ".0.0.0" << endl;
    cout << "Last IP         : " << firstOctet << ".255.255.255" << endl}
  else if (firstOctet == 127) {
    cout << "Class          : Reserved / Loopback" << endl;
    cout << "127.x.x.x is used for troubleshooting." << endl;}
  else if (firstOctet >= 128 && firstOctet <= 191) {
    cout << "Class          : B" << endl;
```

```
    cout << "Subnet Mask     : 255.255.0.0" << endl;
    cout << "No. of Networks  : 2^14 (16,384 possible networks)" << endl;
    cout << "Usable Hosts    : 2^16 - 2 (65,534)" << endl;
    cout << "Starting IP     : " << firstOctet << ".0.0.0" << endl;
    cout << "Last IP         : " << firstOctet << ".255.255.255" << endl;}
  else if (firstOctet >= 192 && firstOctet <= 223) {
    cout << "Class          : C" << endl;
    cout << "Subnet Mask     : 255.255.255.0" << endl;
    cout << "No. of Networks  : 2^21 (2,097,152 possible networks)" << endl;
    cout << "Usable Hosts    : 2^8 - 2 (254)" << endl;
    cout << "Starting IP     : " << firstOctet << ".0.0.0" << endl;
    cout << "Last IP         : " << firstOctet << ".255.255.255" << endl;}
  else if (firstOctet >= 224 && firstOctet <= 239) {
    cout << "Class          : D (Multicast)" << endl;
    cout << "Subnet Mask     : Not applicable" << endl;
    cout << "No. of Networks  : Not applicable" << endl;
    cout << "Usable Hosts    : Not applicable" << endl;
    cout << "Starting IP     : " << firstOctet << ".0.0.0" << endl;
    cout << "Last IP         : " << firstOctet << ".255.255.255" << endl;}
  else if (firstOctet >= 240 && firstOctet <= 255) {
    cout << "Class          : E (Experimental)" << endl;
    cout << "Subnet Mask     : Not applicable" << endl;
    cout << "No. of Networks  : Not applicable" << endl;
    cout << "Usable Hosts    : Not applicable" << endl;
    cout << "Starting IP     : " << firstOctet << ".0.0.0" << endl;
    cout << "Last IP         : " << firstOctet << ".255.255.255" << endl;}
  else {cout << "Invalid IP address." << endl;}
  return 0;}
```

```
--- Results ---
IP Address          : 172.20.1.233
Class               : B
Subnet Mask         : 255.255.0.0
No. of Networks     : 2^14 (16,384 possible networks)
Usable Hosts        : 2^16 - 2 (65,534)
Starting IP         : 172.0.0.0
Last IP             : 172.255.255.255
```

